

Draft - expert review pending

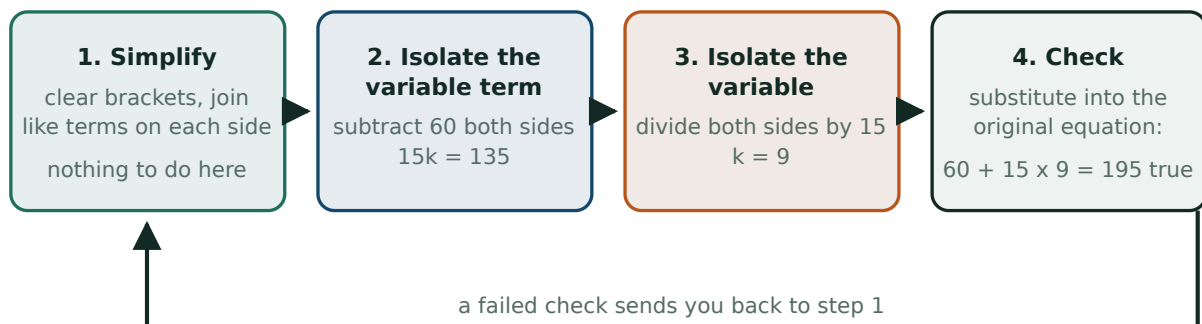
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Topic 3.2 - Linear equations and inequalities

A real classroom situation

Solving a linear equation: the balance method

Example thread: $60 + 15k = 195$



Whatever you do, do it to both sides: that is the whole method.

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A rickshaw driver in Hyderabad posts his rates at the stand: "60 rupees to start, then 15 rupees for every kilometre." A passenger travelling to the station pays 195 rupees and wants to check the fare. The situation hides an equation: $60 + 15k = 195$, where k is the distance in kilometres. The driver solves it without ceremony: "195 minus 60 is 135; 135 divided by 15 is 9; nine kilometres." He has performed the two classic moves - take the fixed part away from the total,

then divide by the rate - and every two-step equation you will ever solve uses exactly those moves, in that order. This topic names the method, adds the safety check the driver skipped, and then widens the idea to inequalities, where the answer is a range rather than a single number.

Explanation

Linear equations

A linear equation claims that two expressions are equal for some value of the variable: $60 + 15k = 195$. Picture a balance: the left pan holds $60 + 15k$, the right holds 195. Whatever you do, you must do to both pans, or the balance lies. That one picture generates every legal move:

- Add or subtract the same amount on both sides.
- Multiply or divide both sides by the same non-zero amount.

The flowchart above turns the picture into an ordered method: simplify, isolate the variable term, isolate the variable, check.

Worked example. Solve $60 + 15k = 195$. Simplify: nothing to do. Isolate the variable term: subtract 60 from both sides, giving $15k = 135$. Isolate the variable: divide both sides by 15, giving $k = 9$. Check: $60 + 15 \times 9 = 60 + 135 = 195$. True. The passenger travelled 9 kilometres. The check is not decoration; it catches the sign slips that silently corrupt fares, bills and budgets.

A common misconception to correct. Many trainees read the equals sign as "the answer comes next" (a habit from primary arithmetic: $2 + 3 = 5$). In an equation, the equals sign is a claim of balance that stays true only under fair moves. Writing $60 + 15k = 195 = 135 = 9$ (a chain seen on many scripts) is the misconception in action: 195 does not equal 135. Keep each step on its own line, and let the equals sign mean balance.

Linear inequalities

Multiplying by a negative turns the number line around



multiply both sides by -2

the whole picture turns around: 5 becomes -10, 6 becomes -12, and "greater than" becomes "less than"



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A linear inequality claims a range, not a point: the school's stationery budget allows at most 2,000 rupees, so $45b \leq 2,000$ for b boxes at 45 rupees. Solving uses the same balance moves with one new rule: **multiplying or dividing by a negative number flips the inequality sign**. The diagram above shows why. If $x > 4$ (everything to the right of 4), then multiplying by -2 turns the whole picture around: $-2x < -8$, everything to the left of -8. The numbers 5, 6, 7 (all greater than 4) become -10, -12, -14 (all less than -8). The flip is not a rule to memorise blindly; it is what the number line does when it is turned around.

Worked example. A school can spend at most 2,000 rupees on geometry boxes costing 45 rupees each, and already owes the supplier 300 rupees from last term, which must be paid from the same money. Write and solve the inequality. Available for boxes: $2,000 - 300 = 1,700$ rupees, so $45b \leq 1,700$. Divide both sides by 45: $b \leq 37.8$ (to one decimal). Since boxes come whole, $b \leq 37$. The school can buy 37 boxes, and the look-back check confirms: $37 \times 45 = 1,665 \leq 1,700$, while $38 \times 45 = 1,710$ would exceed it.

Activity: The fare desk

Work in pairs. Each pair writes three fare-card situations for the other: a fixed charge plus a per-unit rate, with the total paid stated (for example, "200 rupees for a 4-km trip on the 50-plus-25 rate"). The solving pair must write the equation, solve it with the four steps, and check. The writing pair verifies the check, not just the answer. Rotate after three problems. About 20 minutes.

Check your understanding

1. Solve $30 + 12m = 150$ and check your answer by substitution. (*Apply*)
2. Write the equation for: a tutor charges 400 rupees per session plus 250 rupees for materials, and a family paid 2,650 rupees in total. Then find the number of sessions. (*Apply*)
3. Solve $5x - 7 > 18$ and describe the solution as a sentence ("x is greater than ..."). (*Apply*)
4. Solve $-3y \leq 21$, showing the sign flip and explaining in one sentence why it happens. (*Apply*)
5. A trainee writes $60 + 15k = 195 = 135 = 9$ on one line. Identify the misconception in one sentence. (*Analyze*)

Summary

An equation is a balance: legal moves do the same thing to both sides, and the method is simplify, isolate the variable term, isolate the variable, check. The equals sign means balance, not "answer next". An inequality claims a range and uses the same moves, except that multiplying or dividing by a negative flips the sign - the number line turning around. Whole-number contexts (boxes, vans, seats) add a final look-back step: the algebraic answer must be rounded in the direction the situation demands.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can solve a two-step linear equation and prove the answer by substitution.
- ☐ I can translate a fare-or-budget situation into an equation or inequality.

☐ I can solve inequalities, flip the sign correctly on negative multiplication, and explain why.

Try this at your practicum school

Find one posted rate in or near the school (van fare, photocopy rate, canteen price list with a fixed part). Write the equation for a total you or the clerk actually paid, solve it, and check with the person who charged it. Note any disagreement and its cause.

Summative task

A school bus service charges 8,000 rupees per trip plus 120 rupees per pupil. The school's transport budget for one event is 20,000 rupees. In a short written answer: (a) write the cost expression for p pupils, (b) write and solve the inequality for how many pupils the budget can carry, and (c) state the largest whole number of pupils, with the substitution check that proves one more pupil would not fit.

Mini-rubric (each criterion 0-2 marks):

- *Expression:* $8,000 + 120p$.
- *Inequality solved:* $8,000 + 120p \leq 20,000$ gives $p \leq 100$.
- *Largest whole number with check:* 100 pupils, with $8,000 + 120 \times 100 = 20,000$ exactly; the check must show the budget is exactly met (or that 101 would give 20,120 and exceed). **(Analyze)**

Further reading

- Marecek, L., Anthony-Smith, M., & Mathis, M. H. (2020). *Prealgebra 2e* (Ch. 8 Solving Linear Equations). OpenStax.
<https://openstax.org/details/books/prealgebra-2e>
- National Curriculum Council, Ministry of Federal Education and Professional Training, Government of Pakistan. *Mathematics: Suggested Guidelines (Grades 1-8)*.

<https://ncc.gov.pk/Detail/ZjgzYzg2MmMtZDc0Zi00NjEzLTk5ZmYtZGJiNjc5ODljOGUx>

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